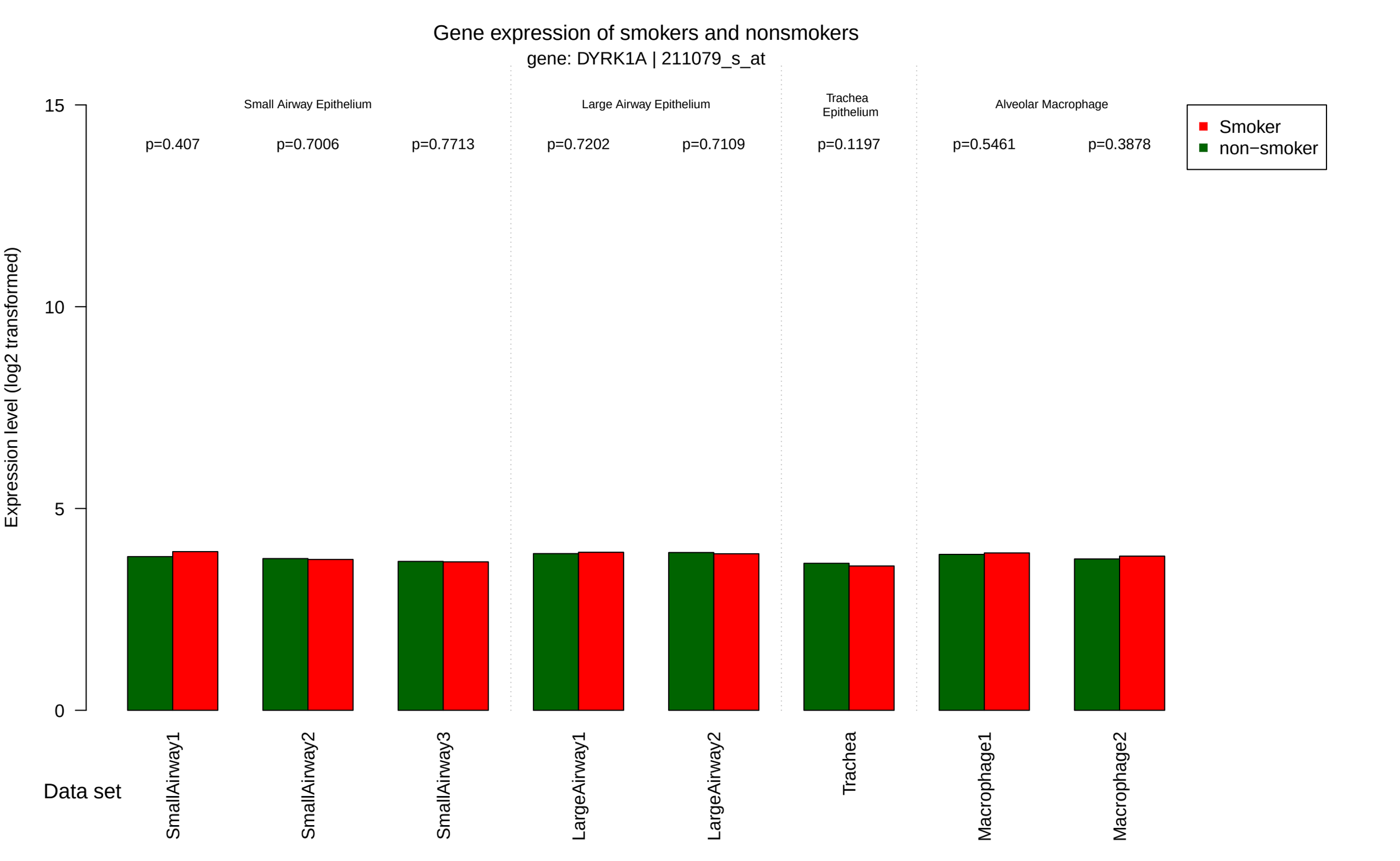
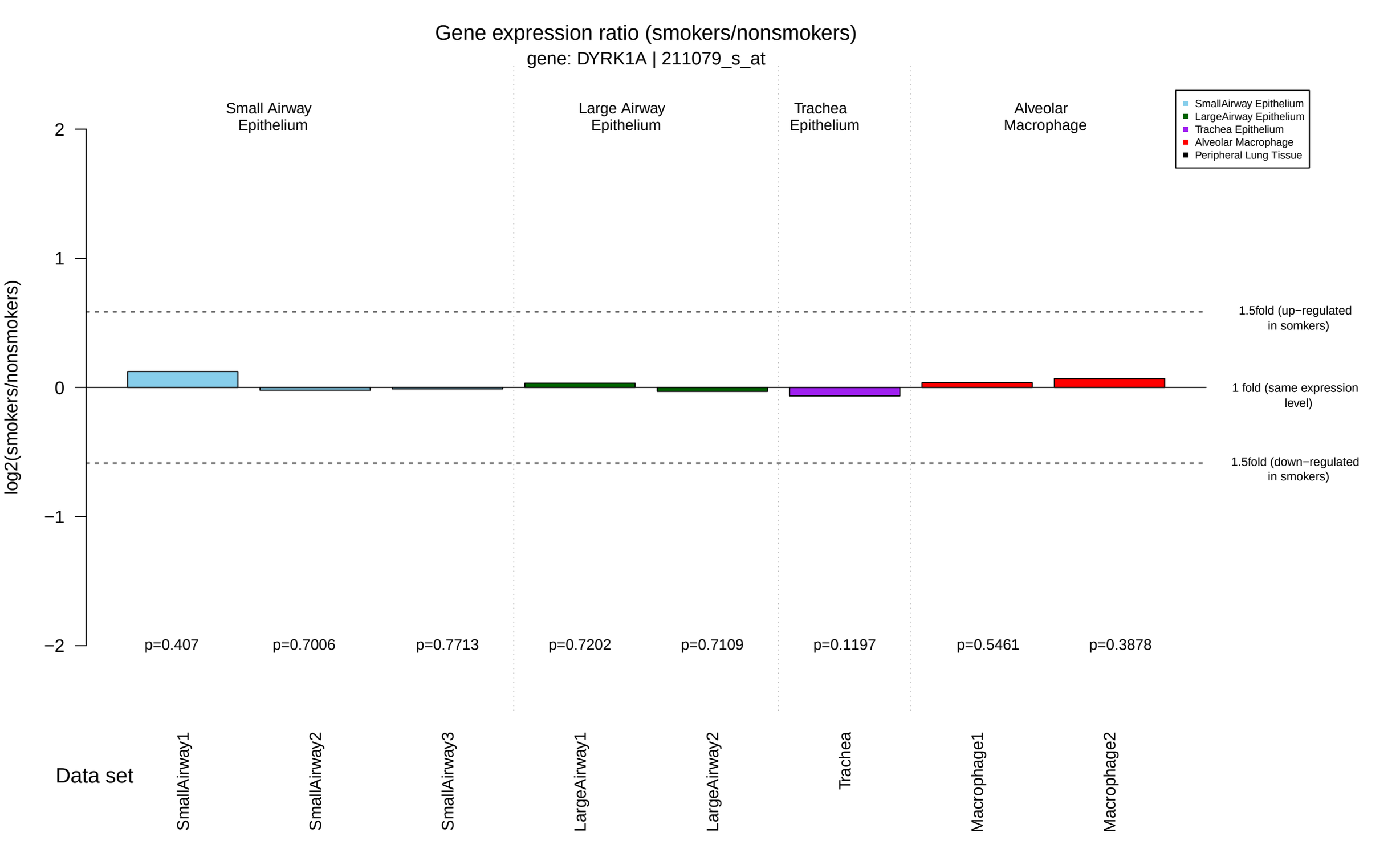
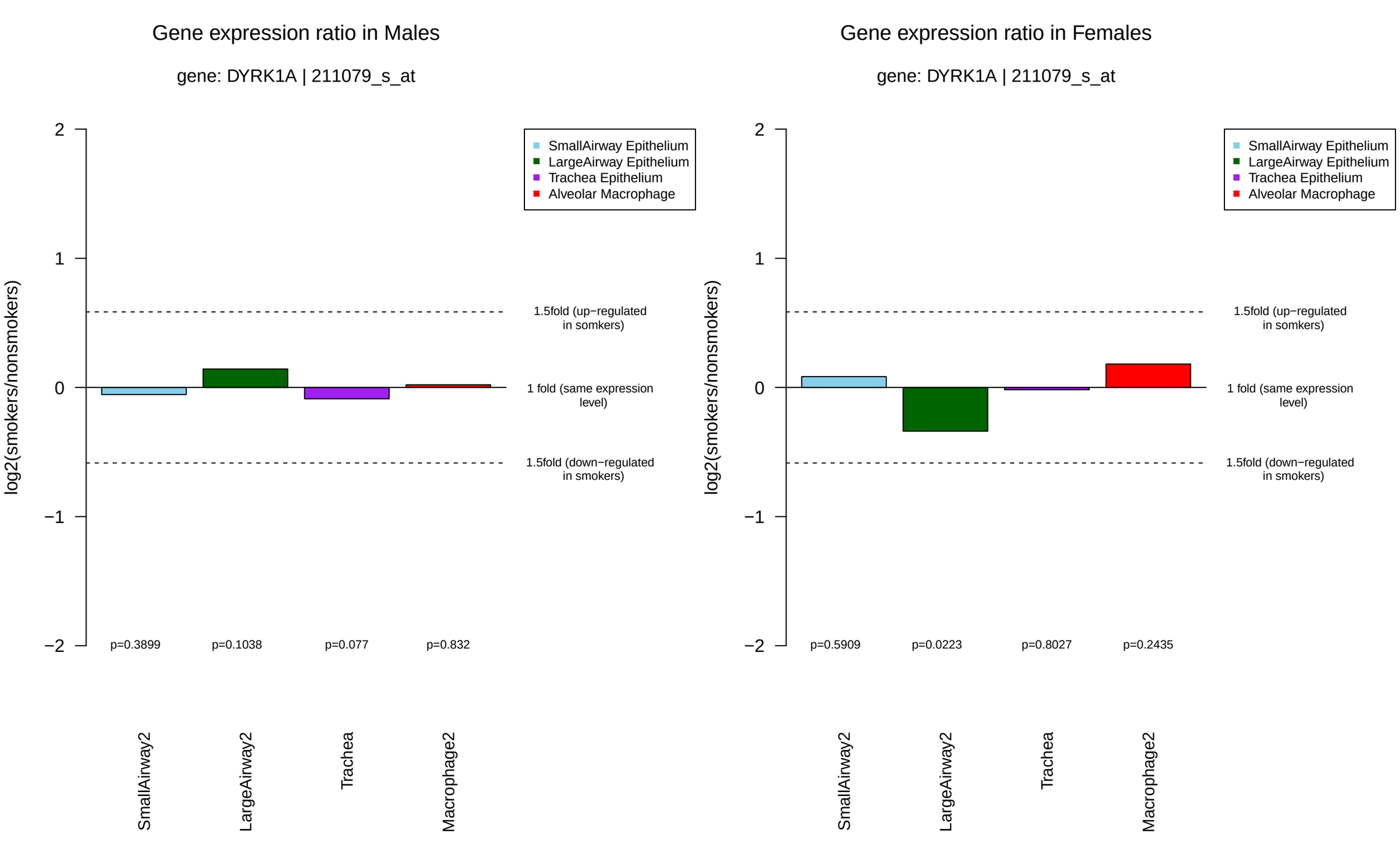
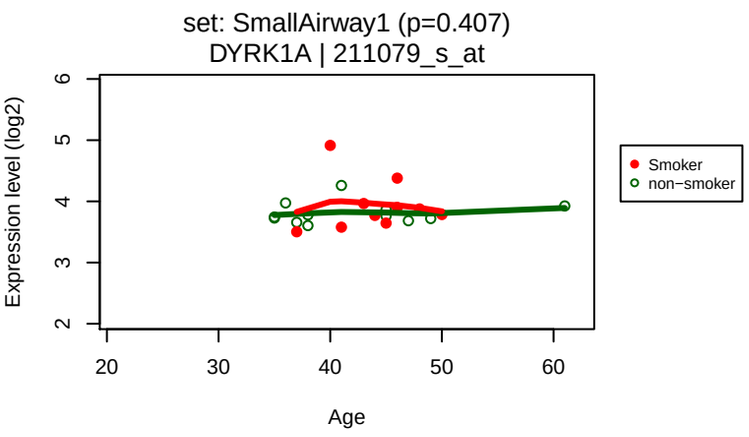
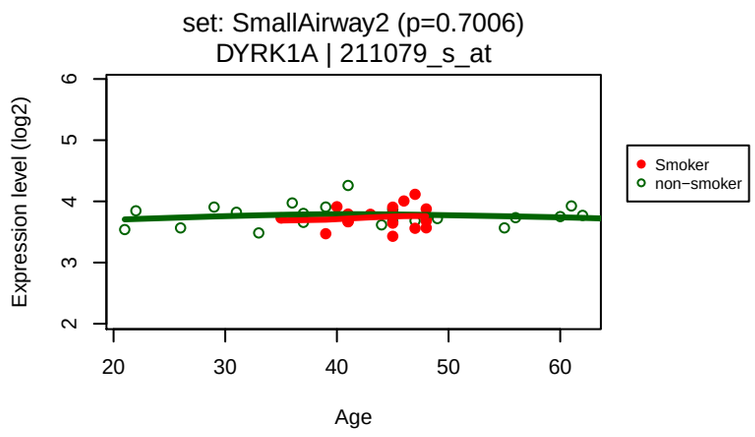
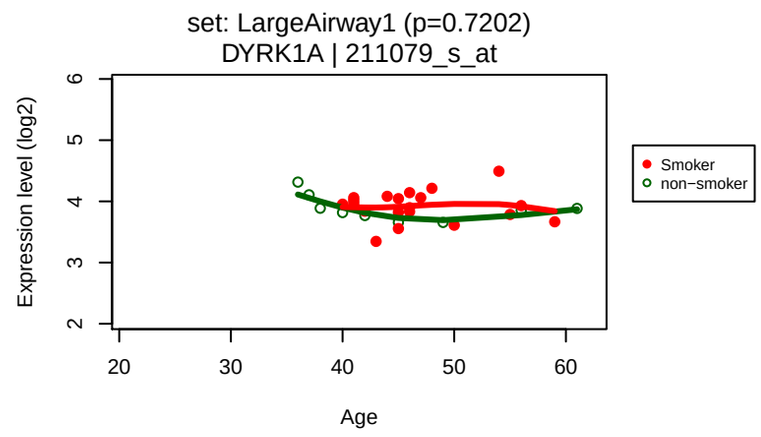

Gene expression of smokers and nonsmokers
gene: DYRK1A | 211079_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.407
p=0.7006
p=0.7713
p=0.7202
p=0.7109
p=0.1197
p=0.5461
p=0.3878
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

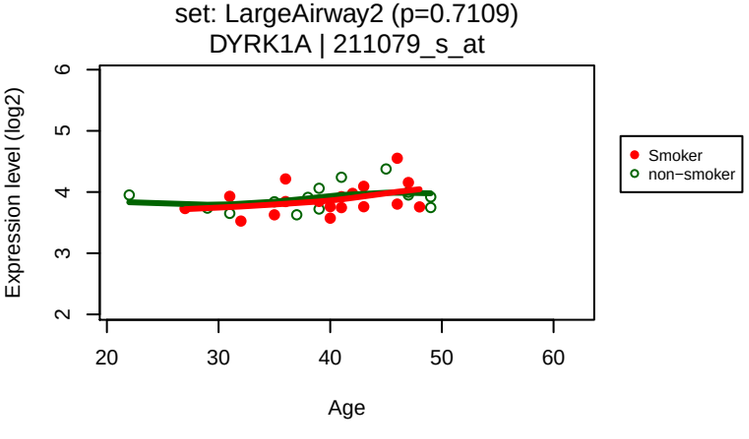

Gene expression ratio (smokers/nonsmokers)
gene: DYRK1A | 211079_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.407
p=0.7006
p=0.7713
p=0.7202
p=0.7109
p=0.1197
p=0.5461
p=0.3878
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

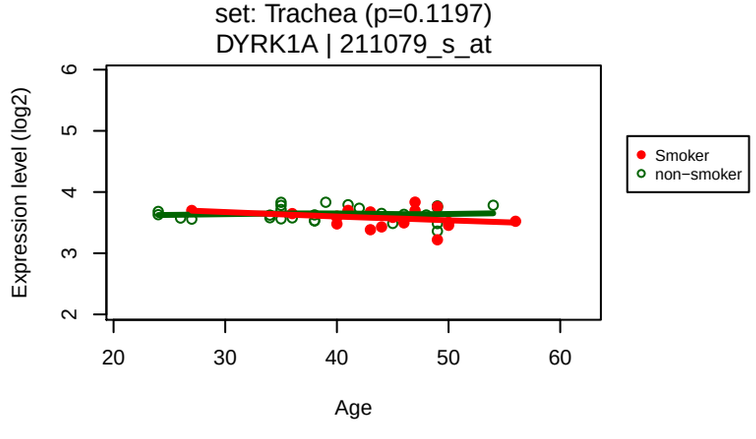

Gene expression ratio in Males
gene: DYRK1A | 211079_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3899
p=0.1038
p=0.077
p=0.832
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: DYRK1A | 211079_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5909
p=0.0223
p=0.8027
p=0.2435
SmallAirway2
LargeAirway2
Trachea
Macrophage2

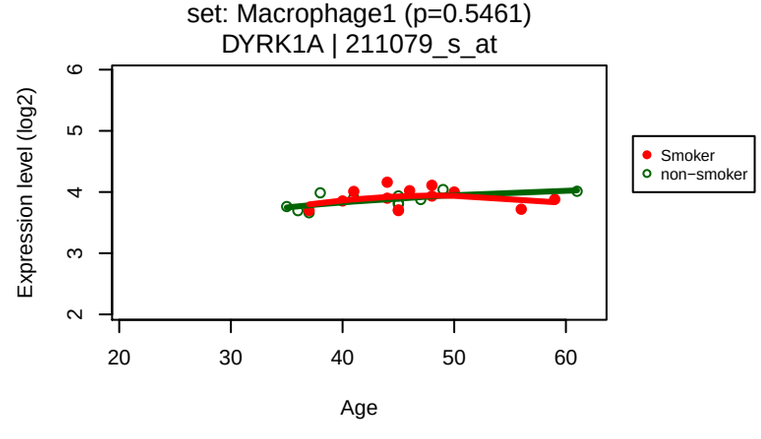

set: SmallAirway1 (p=0.407)
DYRK1A | 211079_s_at
Expression level (log2)
Age
Smoker
non−smoker

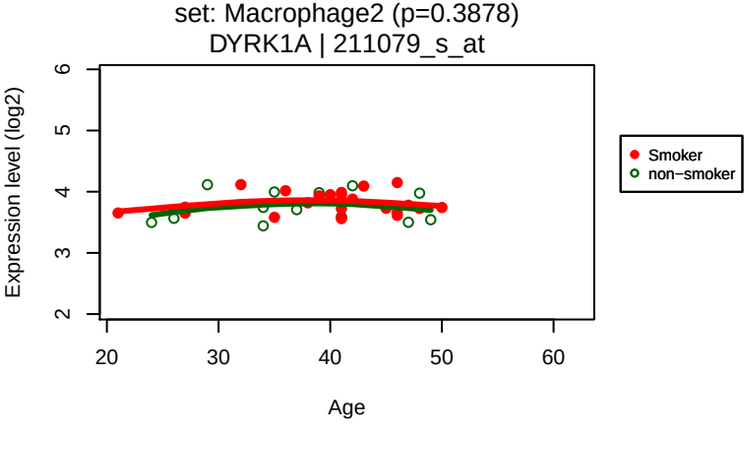

set: SmallAirway2 (p=0.7006)
DYRK1A | 211079_s_at
Expression level (log2)
Age
Smoker
non−smoker

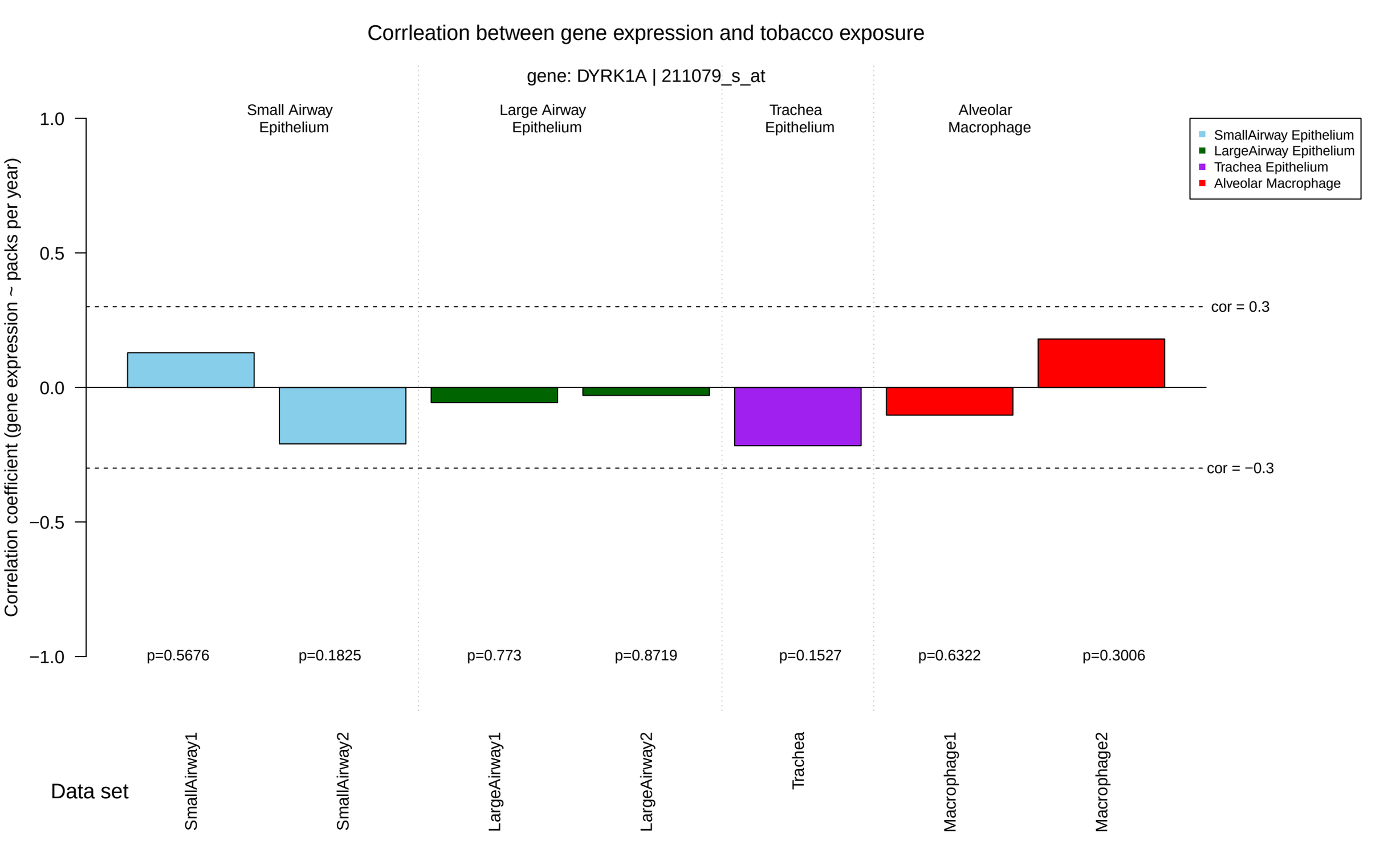

set: LargeAirway1 (p=0.7202)
DYRK1A | 211079_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7109)
DYRK1A | 211079_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1197)
DYRK1A | 211079_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5461)
DYRK1A | 211079_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3878)
DYRK1A | 211079_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: DYRK1A | 211079_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5676
p=0.1825
p=0.773
p=0.8719
p=0.1527
p=0.6322
p=0.3006
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2